

The 7400 series are transistor–transistor logic (TTL) logic chips. First introduced in 1966, they still find application in digital circuits. One of the members of the family is the 74LS00, which is described as "quad 2-input NAND gate". We would like to test the functionality of the 74LS00 by physically constructing a circuit to observe the NAND output for all possible input combinations.

The circuit to be constructed is visualized in Fig. 1.

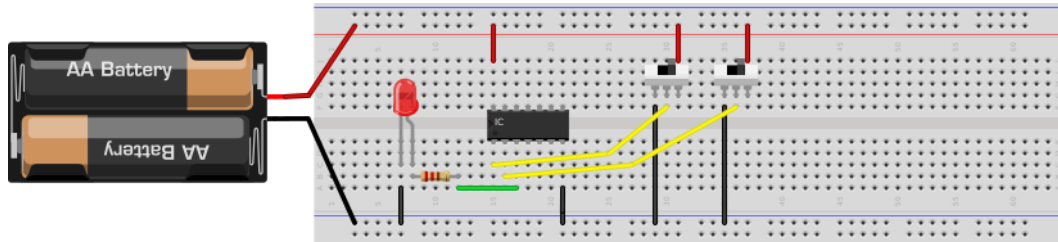


Fig. 1: The 74LS00 test circuit on a breadboard.

The expected behavior of the circuit can be observed by a *Digital* simulation. A screenshot of 74ls00.dig (available on SUCourse) is shown in Fig 2.

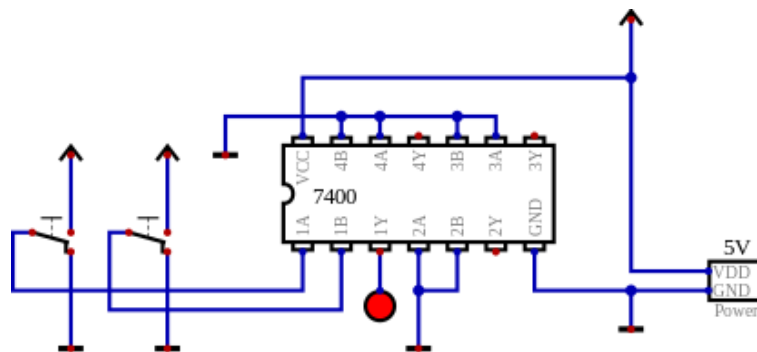


Fig. 2: The 74LS00 test circuit in *Digital*.

During the laboratory session, you will construct and test this circuit. You will be then asked to make a modification to the circuit to implement a different function. You may work in groups of two.

Before the laboratory session,

1. Read the datasheet of 74LS00.
2. Learn how a breadboard is used.

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